

# Automatic Street Light Using LDR Sensor

**Objective:** To control an LED automatically based on light intensity using an LDR sensor and Arduino.

**Components Used:** Arduino Uno, LDR Sensor, LED, 220 $\Omega$  Resistor, 10k $\Omega$  Resistor, Breadboard, Jumper Wires.

**Working Principle:** The LDR senses surrounding light intensity. Arduino reads the sensor value through analog pin A0. When light intensity decreases, the LED turns ON. When light intensity increases, the LED turns OFF.

**Code Explanation:** Arduino continuously reads the LDR value using `analogRead(A0)`. The value is compared with a threshold and the LED connected to pin D3 is switched ON or OFF accordingly.

**Conclusion:** This project demonstrates a simple sensor-based IoT simulation for automatic street light control using an LDR sensor and Arduino.